

LLM Enhanced Threat Intel Correlation Automation using n8n

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Abstract. In today's cybersecurity practice, real-time collection, validation, and action on threat intelligence is paramount to effectively reducing risk to the organization. However, conventional threat intelligence processes tend to be disjointed, manual or constrained to static data feeds from limited sources, resulting outdated threat intelligence which leads to delayed responses to new threats. The proposed solution presents a completely automated threat intelligence and reporting generation pipeline based on n8n, incorporating Perplexity LLM for smart open-source intelligence (OSINT) gathering and Telegram as a means for real-time analyst engagement. The entire intelligence lifecycle is fully automated with alerts through Telegram, retrieving the most recent threat reports specific to products or versions with the use of LLM queries, parsing out Indicators of Compromise (IOCs) including CVEs, IPs, domains, and hashes, and enhancing them leveraging APIs of VirusTotal, AbuseIPDB, and HybridAnalysis. An impact classifying logic then produces natural language analytical reports with structure, which are instantly sent to the analyst. Exploiting the reasoning power of large language models and automating enrichment processes, this framework greatly improves throughput, accuracy and contextual richness of threat intelligence, minimizing reliance on static feeds or manual triage.

Keywords. Threat Intelligence Automation, Large Language Models, Perplexity LLM, Open-Source Intelligence, IOC Enrichment, VirusTotal, AbuseIPDB, Hybrid Analysis, Telegram Bot, n8n Workflow, Real-Time Reporting

INTRODUCTION

The proliferation of digital ecosystems, integrated systems and networks has necessitated threat intelligence to be a part of today's cyber operations. Enterprises struggle to sift through the sheer volume of security data they receive today from a multitude of open-source intelligence (OSINT) feeds, advisories, and vendor disclosures. The problem is how to quickly aggregate, validate, and analyze that information in real time. Most traditional corporate environments rely on few static or vendor-specific intelligence feeds which lead stale information and reactive responses to new threats. Now that cyber adversaries are constantly evolving their tactics, techniques and procedures (TTPs) it is apparent that traditional threat intelligence techniques are insufficient for providing timely or actionable insights.

The response, already complicated by the lack of automation for collecting and enriching threat indicators. Analysts are burdened by tedious and manual work like finding vulnerabilities, extracting Indicators of Compromise (IOCs) and matching them across different databases. This not only decides slower but also makes it possible to miss essential intelligence. To overcome these constraints, our framework proposes a novel hybrid end-to-end threat intelligence and report writing pipeline utilizing LLMs for reasoning, coupled with powerful enrichment and delivery mechanisms.

The proposed approach built on n8n, an automation platform, incorporates the Perplexity LLM for on-the-fly gathering and analyzing of OSINT threat intelligence. By submitting vendor- and version-specific input lists to the LLM, the framework extracts information about the current vulnerabilities, exploits, and active threat indicators. FBI IOCs such as CVEs, IPs, domains or hashes are automatically enriched via APIs with VirusTotal, AbuseIPDB and Hybrid Analysis. A severity categorization mechanism to evaluate the fused enrichment results for the overall risk associated to the collected intelligence.

A. Automated Data Collection and Trigger Mechanism - The system is grounded by a Telegram-based trigger that users can customize by entering keywords including product names or software versions. These inputs are running through a chain of automation processes with n8n where the Perplexity LLM is doing focused OSINT queries to gather recent information from trusted sources. Employing a conversational UI makes the interaction natural and security analysts-friendly.

B. IOC Extraction and Enrichment Workflow - After fetching the data, the framework then applies text processing techniques on the relevant URLs and documents to extract IOCs. Then each IOC is checked for potentially malicious content by multiple enrichment queries with VirusTotal, AbuseIPDB and Hybrid Analysis APIs. The enrichment step groups the results for a severity assessment.

C. Threat Classification and Report Generation - The enrichment information is run through a severity voting logic that evaluates overall threat level to LOW, MEDIUM and HIGH based on average scores (across different sources). The framework then produces a complete analytical report which includes an executive summary, IOCs, enrichment results and available security mitigations. The end report is in real time delivered to the user through Telegram or optionally through email.

Future development of this system may include integration with dynamic data feeds, adaptive LLM prompting to enhance contextual accuracy, as well as integration with security monitoring solutions like SIEM or SOAR systems for orchestrated incident response. By streamlining and enriching the intelligence lifecycle via automation, this framework minimizes manual effort, enhances situational awareness, and expedites detection and response in dynamic threat environments.

LITERATURE REVIEW

The literature review presented here incorporates many different studies and research papers that deal with automation of Cyber Threat Intelligence (CTI), Automated Threat Analysis using Large Language Models (LLM) and Integration of Enrichment Tools for Security Operations, which have seen rapidly changing technologies including automated extraction, reasoning and automation of threat data, while at the same time identifies current limitations in both the ability to perform real-time orchestration, as well as the operational efficiency of deployments based on these technologies.

CTI-Bench [1] was created as a benchmark to evaluate the capabilities of LLMs to perform CTI related tasks (on extraction of indicators of compromise (IoCs) and the reasoning process to arrive at a decision). While this benchmark has been shown to provide an effective evaluation of the overall capabilities of the models used in the test environment, the creation of CTI-Bench [1] does not adequately consider many of the operational characteristics of the application of LLMs in this environment including speed (latency) and real-time data integration, as well as handling of dynamic open-source security threat intelligence (OSINT) feeds. A related study [2] introduced a fully automated end-to-end a CTI-to-Incident Response system which takes a set of extracted TTPs, and translates them into response actions. This system did not offer a scalable application of LLM- automated dynamic retrieval or integration of live OSINT data sources.

The paper in [3] studied the application of the VirusTotal API to help with the triaging of alerts in a security operations centre, as well as showing how to use workflow enrichment to minimise false-positive count rates. However, the research undertaken is limited in scope and does not take into consideration any rate-limiting, privacy, or full pipeline automation issues that may arise. Furthermore, the automated evaluation of open-source intelligence (OSINT) sources based on feature extracts such as freshness and location; has also been proposed in [4]. Although this is an interesting proposition, it does have limitations when faced with either non-English/non- Western language or adversarial web content that are often present in practical cyber threat intelligence (CTI) use cases.

The study in [5] considered the use of artificial intelligence (AI) based annotation tools to enable cyber defence data to be labelled and structured into a format that is suitable for automated processing. Despite their benefits, the inability to track and mitigate annotation drift and to stop the propagation of errors were significant obstacles. In contrast, [6] presented a new way of constructing CTI knowledge graphs with an LLM- based framework called CTINexus by extracting both entities and their relationships. While innovative, this approach showed signs of hallucination, poor entity mapping and issues with realtime streaming based updates. Hence, [7] presented an LLM-based proactive CTI

model called GRA856 (Grounded Retrieval-Augmented Generation), which takes real-time CVE and EPSS feeds and uses them to infer emerging threats. While the GRA856 model is definitely promising, it does have issues with being able to ascertain the reliability of sourcing and also has a significant inference cost and latency when working at the large size. A review and analysis of similar work in cyber threat intelligence is provided in [8].

The article referenced as [7] describes a proactive CTI model using LLMs. In this model, retrieval-augmented generation (RAG) and real-time CVE and EPSS feeds are combined to predict possible upcoming threats. The model appears to hold promise; however, there were some challenges that were encountered with regards to its reliability as well as the cost and latency associated with inferences made with it on an operational, or large scale, basis.

The article [8] is a general overview of the different uses of LLMs in the cybersecurity context, such as IOC extraction and summarization. This article does not include an empirical assessment of the integration of LLMs into automation platforms, such as n8n or real-time APIs (Perplexity). In [9] is a comparative study of rule-based versus LLMs for the extraction of IOCs, and it was found that LLMs had a higher overall completeness but that both hallucination and token limits resulted in significantly lower precision. The article [10] provides a general overview of the cybersecurity landscape, including the most common attack and countermeasure surfaces, however, it does not provide any concrete frameworks for either the automation of these measures or for unified threat intelligence workflows.

The article [11] discusses new advancements made in the area of Cybersecurity Knowledge Modeling, specifically with the use of ontology-based reasoning in AI-enabled cyber defenses. However, the findings of this article are limited because the ontologies were primarily based on manually curated content; there was no mechanism for automatically taking in dynamic feeds from CTI sources. Finally, the paper in [12] investigates the general movement toward either low-code or no-code platform usage for all industries—but it does not specifically include the area of Cybersecurity Automation, nor does it include any CTI orchestration, which suggests an opportunity to integrate with platforms similar to n8n.

The previously mentioned papers reviewed the literature available on retrieval-augmented architectures [13], and while they described approaches to enhance the contextual recall from LLM's through external data retrieval, they did not include specific examples of how to ingest OSINT feeds or extract features from HTML. In a similar manner, [14] focused on novel methods using Natural Language Processing, Transformers, and integration of enhanced Indicators of Compromise (IOC) using third-party resources such as VirusTotal or AbuseIPDB.

AI-based OSINT gathering and classification techniques were also examined in [15], and there was evidence that this type of work can be performed on a significantly larger scale in CTI data collection; however, no standardised, deployable architecture exists for enriching and reporting the results of these techniques. In the TIPCE report [16], a comprehensive analysis of threat intelligence toolset comprehensiveness over time was presented, and while the report did provide some recommendations for developing automation strategies for continuously updating threat intelligence tools, it did not provide any direction.

In [17], paper on Generative Information Extraction Models using LLM's to produce structured outputs was conducted, however the methods presented for generative information extraction models were not modified to incorporate IOC- oriented or operation intelligence pipelines. The final article reviewed in this assessment was [18], which introduced URLBERT; a Transformer-based model specifically developed for the classification of adversarial URLs. URLBERT demonstrated high levels of accuracy when used for static classification but did not possess dynamic capabilities that would allow it to provide real-time enrichment or synthesis of reports.

To summarize, the body of literature highlights five major gaps in the current paper related to the following: (1) there is no automated system or pipeline that fully integrates LLM reasoning into IOC-based enrichment; (2) there are limited workflows available for generating real time intelligence that support the ability for adaptive development against changing threat patterns over time; (3); there are no quality systems in place to provide low latency report generation that are also reliable; (4) very few studies have evaluated or addressed the problems of hallucination or grounding of language models within the operational sector; (5) very few studies have focused on the deployment of lightweight, low-code technologies in Security Operations Centers (SOC).

Based on these limitations, the proposed framework integrates LLM-based OSINT research and dynamic IOC gathering and development through the combined use of multiple API services (VirusTotal, AbuseIPDB, HybridAnalysis) and automated report generation and delivery workflow (managed through n8n and Telegram). We're

bringing together two really different research fields. Our new way of doing things fills a lot of holes in current papers and gives us a structure that lets us build models that can grow easily. Plus, it's easy to understand and straightforward to put into action for automatically getting cyber threat intelligence in real-time.

PROPOSED IDEA

This paper presents a framework that automates the collection, augmentation, and sharing of cyber threat data as new threats emerge. This framework combines an LLM with the n8n automation platform to collect threat data automatically from available API sources, add related context from other API-based tools, and produce a threat report with minimal manual work. This automated framework is intended to replace the current manual workflow in the Security Operations Centre (SOC), reducing process inefficiencies and shortening the time needed to respond to incidents [1],[2]. The process starts when a user searches for details about a cyber threat, either by asking a direct question or by entering a trigger term such as a malware name, a CVE identifier, or a threat actor name, through the Telegram interface.

This query starts an automated n8n workflow that collects data through the Perplexity LLM API. The retrieved data includes real-time OSINT on threats from reputable cybersecurity data sources, along with contextual details such as IOCs and the malicious infrastructure linked to the threat [3], [4].

After data collection is finished, the next module processes the dataset and automatically extracts any IOC(s). After the LLM produces its output, custom parsing code extracts the indicators of compromise and sorts them into four types: IP addresses, URLs, domains, and file hashes. These groups are then split into smaller batches so they can be processed in parallel during the enrichment step. During the enrichment phase, the system queries several public threat-intelligence APIs, such as VirusTotal, AbuseIPDB, and HybridAnalysis, to check each IOC, compare findings across sources, and assign a risk score [5], [6]. VirusTotal scans files and URLs for malware by checking them with many different engines. AbuseIPDB gathers user reports about harmful or suspicious activity linked to specific IP addresses. HybridAnalysis checks whether a file or URL may be malicious by studying how it behaves during analysis. The IOCs are then gathered into a single dataset that includes risk scores, last-seen dates, and vendor consensus.

This dataset is collected and then converted into an automated threat report, produced in either Markdown or PDF, to support sharing and long-term storage. The automated reports are sent to users through the Telegram messaging app via an n8n workflow, giving them an almost immediate summary of the intelligence [7], [8]. Figure 1 shows an event-driven pipeline architecture designed to scale, reduce manual work through automation, and remain easy to interpret.

The framework or profile is made up of separate modules that allow workflow nodes to run as needed during execution. The system can be expanded over time by adding extra enrichment APIs and new data sources as the threat landscape changes. To improve accuracy and limit made-up content in responses from large language models, the framework applies post-processing checks that use regex patterns to confirm the extracted fields and then compares them against enrichment sources to see if they agree. This helps keep out false or unverified information and makes the results more reliable [10], [11]. The enrichment layer supports IOC correlation by flagging shared infrastructure and recurring indicators across several threat reports.

This low-code, cost-efficient approach supports building complex integrations that combine APIs and machine learning tools using n8n. This method of building workflows enables API Integrations that otherwise would require significant effort to develop from the ground up using traditional backend programming languages [12].

This modularity of the n8n workflow allows for quick changes between API keys across various systems and also allows you to easily enrich nodes and update LLM endpoints, thereby making it a great tool for small academic research environments or SOC's deploying on a small scale. The reporting module is designed to produce structured reports, these reports consist of automated reporting outputs, including contextual summaries, IOC tables and enrichment statistics, as well as the ability for the users to generate their own structured reports (including downloading and scheduling) and provide continued monitoring of active campaigns or emerging threats.

TABLE 1. *Configuration Specifications Of Proposed Framework*

Component	Function	Integration Tool
Telegram Node	User query input and report delivery	Telegram Bot API
Perplexity Node	Threat data retrieval and contextual search	Perplexity LLM API
IOC Extraction Node	Regex-based parsing and type segregation	Custom Script
VirusTotal Node	URL/IP/Hash enrichment	VirusTotal API
AbuseIPDB Node	IP reputation verification	AbuseIPDB API
HybridAnalysis Node	Behavioral threat analysis	Hybrid Analysis API

The table identified as Table1 lists the various configuration parameters that have been used to develop the workflows outlined in this paper, which lists all configuration parameters as well as the API endpoint and node function used in developing the workflow.

The nearly real-time workflow is designed to deliver full enrichment and reporting (i.e., generating a report of suspicious activity) within 20-40 seconds based on how many queries were submitted and how long it took for responses from the API to be returned to the end-user. The automation enables SOC staff to do higher value processing of alerts (e.g.. Triage), and conduct all the activities that are proactively searching (e.g.. hunting) for cyber threats in a timely manner (e.g.. Responding and alerting).

The integrated automated architecture as described this paper specifically bridges three critical gaps in traditional threat intelligence processes - 1) Real-time synthesis of multiple feed types; 2) disjointed processes for enriching the input data; and 3) limited automated report generation for intelligence gathering.

It also allows for a more scalable and modular approach to the management of a dynamic cyber threat intelligence operation as well as aligning to modern security orchestration principles and AI-driven intelligence enrichment methods [13], [14], [15].

SYSTEM DESIGN

The ITEF-based threat intelligence automation framework is proposed and depicted in Figure1, which is a staged workflow form for collection, enrichment and automated reporting of data. Leveraging LLM-assisted threat intelligence analysis, IOC extraction, and multi-API enrichment, the platform provides a unified and scalable architecture for real-time analysis and sharing of cyber threats [1], [2].

A.Threat Query and Data Retrieval - The first phase of the framework starts with a user query, which can be submitted manually through a Telegram bot interface or by automated scheduling in n8n. When a user submits a keyword, e.g., a malware family, a threat actor, or a CVE identifier, the query is sent to the Perplexity LLM API for contextual OSINT searching and summarization. Rather than relying on static threat feeds, this LLM-based query searches on- demand for up-to-date, pertinent information using open and reliable cybersecurity sources [3], [4].

Thus, the collected intelligence is not limited to static, predefined databases, but may also incorporate information derived from threat reports, advisories, as well as disclosures originating from the community. The Perplexity API uses a secure bearer token for the authentication and the telegram bot is the Bot API for talking to telegram in realtime. The output data provided by the LLM are textual summaries, context, and indicators of threat embedded in the text.

For more information on the n8n node, please see here. Sensitive values such as API tokens or bot credentials are stored securely in the n8n credential store To enable operational security-sensitive such as API tokens and bot credentials are stored securely in the n8n credential manager. The modular workflow architecture enables parallel processing of queries and retrieval of data on demand or at scheduled time intervals to provide continuous threat monitoring and intelligence gathering.

B.IOC Extraction and Preprocessing - After receiving the LLM output, the framework continues with ioc extraction and preprocessing. This phase uses regex parsing in conjunction with text segmentation to extract indicators

of compromise (IOC) such as IP addresses, domain names, URLs, file hashes [5]. Extracted IOCs are classified and verified to eliminate duplications and false matches.

The preprocessing also normalizes IOC formats (e.g., URL encoding forms, domain name normalizations) to be compatible with external enrichment APIs. The extracted indicators are then categorized by type and submitted as structured JSON documents for further processing. This modular extraction pipeline permits (i) high throughput, and (ii) system adaptation to different input formats such as textual reports, JSON objects, or HTML documents collected from OSINT resources [6].

The framework takes advantage of n8n's workflow concurrency to parallelize data cleaning and extraction, so you can respond quickly, even processing large OSINT datasets. This stage is the basis for correct enrichment and stops errors being introduced in risk scoring. Enrichment and External Intelligence Integration

C. Enrichment and External Intelligence Integration - Following the extraction of IOC, the system automatically enriches the IOC by querying multiple threat intelligence providers to determine the risk level and reputation of each IOC. Three main APIs are integrated into: VirusTotal, AbuseIPDB, and HybridAnalysis.

The VirusTotal API checks file hashes, URLs, and domains against multiple antivirus and sandbox engines. AbuseIPDB Access the wealth of information on IP address abuse reports for detecting known malicious or compromised hosts. HybridAnalysis is a fully automated system for the analysis of files and URLs employing sandbox technology to obtain additional information about malware samples or suspicious domains [7], [8].

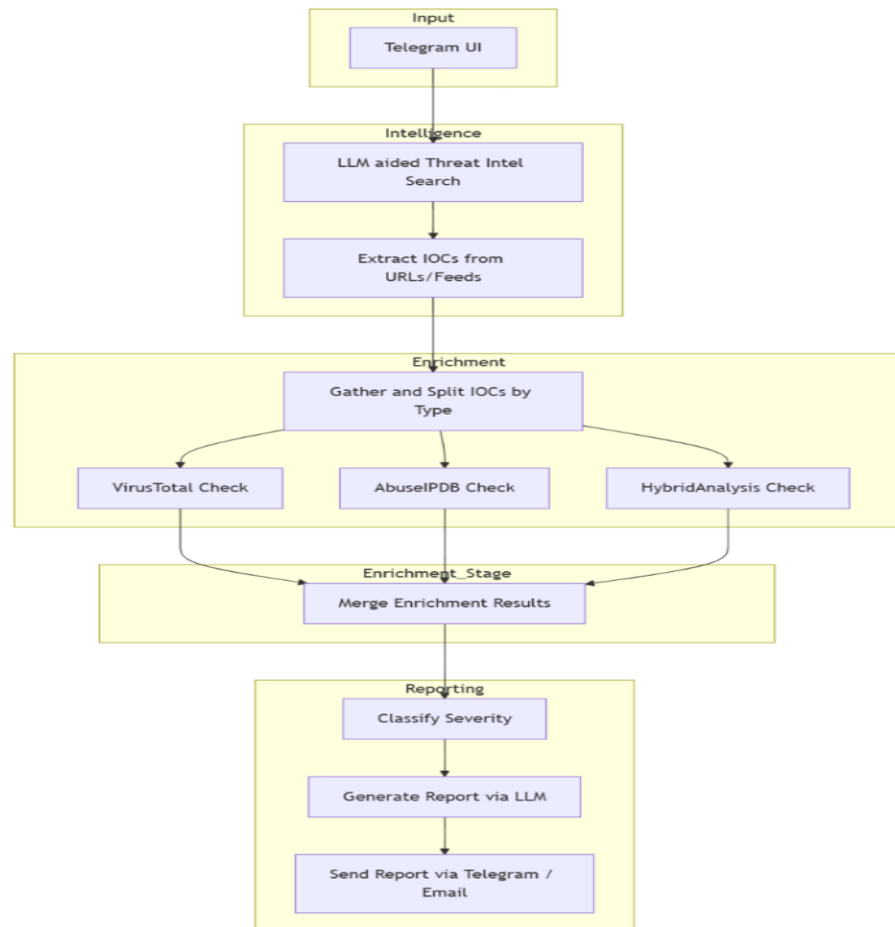


FIGURE 1. Proposed framework architecture

As represented in Figure 1 each enrichment is executed in parallel on separate n8n nodes and the results are joined together in a single re-port. The response data of these APIs are parsed, formatted and mapped into the respective IOC, creating a structured enrichment table with fields like last-seen date, threat confidence, reputation score and number of detections. This enrichment process also promotes the believability of the information as it is cross checked with other reliable sources. It also guarantees that false-positives produced from LLM outputs or noisy OSINT information are reduced by means of reputation- based correlation [9], [10].

Besides enrichment, the framework also uses a rule-based threat severity calculation to split indicators into High, Medium or Low risk categories. The intention of this scoring logic is that it should be treated as an indicator of risk and not an absolute decision, so that analysts can prioritize threats quickly while maintaining their ability to interpret the score.

Severity rating is calculated based on the combination of CVE severity scores and multi-source IOC validation results. If an extracted Ioc is linked to a known CVE, the CVSS severity score is taken as a reference. Indicators that are associated with high or critical CVSS-rated vulnerabilities directly contribute to a higher risk rating.

For indicators not based on CVE or for nascent threats, the severity level is assigned by following a process of cross-source verification. If the IOC is looked as malicious by at least two enrichment sources among (VirusTotal, AbuseIPDB, HybridAnalysis), then the indicator will be considered as High risk, indicating a strong agreement about this indicator among different intelligence providers. Indicators confirmed malicious from a single source are considered a Medium risk as these likely represent threat activity but require further verification by analysts. If an indicator has not been flagged as malicious but historical enrichment shows that it was malicious in the past and has been remediated or otherwise mitigated, the indicator is considered to be Low risk. This methodology allows for previously compromised (now cleaned) systems to be accounted for, without generating excessive alerts and still retaining historical context. The multi-factor severity calculation model achieves a good risk balance by associating the severity of the vulnerability, the consensus in the enrichment, and the behavior of the historical indicator which limits false positives and enhances the quality of the decisions.

D. Threat Report Generation and Delivery - Once the data is enriched, the last part is to aggregate and create the threat intelligence report. The report contains the original query context, a summary of extracted and validated indicators, enrichment results, and overall threat insights. A Markdown report is initially created inside n8n which is then converted to a PDF document that can be downloaded (using a document conversion node [11]).

The final report also includes the derived threat severity level as a summary risk classification. Each report clearly states the threat posture worthiness if High, Medium or Low with a short explanation of what factors (CVE severity scores, multiple source malicious detection, etc.) contributed to the rating. This level of transparency allows analysts to know not only the final risk level, but the rationale behind it, enabling better-informed incident response and prioritization.

The produced report has traceability in the form of entries with timestamps and it can be sent to users in the form of Telegram messages or stored in a local or cloud repository. Reports can be scheduled for repeating delivery, enabling organizations to keep intelligence logs up to date with no manual intervention. This automated reporting channel guarantees rapid delivery of actionable intelligence to security analysts, incident responders and SOC teams. In addition, the framework allows for alerting when high-risk indicators are found so that threat responders can react immediately.

E. Workflow Architecture and Security Considerations - The entire process can be divided into modules, as shown in Figure1, and follows a low-code approach. Each phase query retrieval, LLM search, IOC extraction, enrichment, and reporting is a separate node in n8n, providing fault isolation and scalability. Fallback methods guarantee a continuous process flow in the event of rate limit or errors (API) [12]. To ensure data privacy and compliancy, all node-to-node communication is encrypted, and only the minimum amount of information necessary is logged for audits. No sensitive user data is persisted beyond the session-based cache. The modular architecture also allows future enhancements such as integration with MITRE ATTCK mapping, additional enrichment sources, or AI-based risk scoring.

When LLM-based intelligence, an auto-enrichment process, and reporting during pipeline execution are sandwiched into a single form of well-defined orchestration, the resulting system architecture can provide an end-to-end solution

available for dynamic cyber threat intelligence automation. It reduces the workload of analyst, shortens the delay for sharing intelligence, and achieves perpetual situational awareness in dynamic threat environments [13], [14], [15].

RESULTS AND DISCUSSION

The described threat intelligence automation flow was implemented successfully in the environment of n8n workflow automation. The whole architecture is build up with networked nodes that have different but dependent roles such as receiving user input from telegram, generating report and sending out reports. Every workflow module is run sequentially and automatically, enabling a full threat intelligence life cycle from collection to actionable reporting.

Figure 2 shows the n8n implementation of the pipeline. The architecture consists of several functional nodes, the Telegram Trigger, the Perplexity LLM Search, IOC Extraction, IOC Enrichment, Report Generation and Telegram Delivery. The modular construction allows for flexibility, isolation of errors and for scalability to include future enrichment and/or alerting tools.

The Telegram Trigger node is the entry point for input where an analyst submits a query with product name and version number (e.g. "Fortinet 7.4" or "VMware ESXi 8.0"). The following nodes in the workflow then parse that input, run contextual OSINT queries through the Perplexity API, extract indicators of compromise (IOCs) from the retrieved threat reports, and enrich them via external APIs like VirusTotal, AbuseIPDB, and Hybrid Analysis. The end result is an automatically generated threat intelligence report that is organized and formatted for consumption by an analyst without additional work.

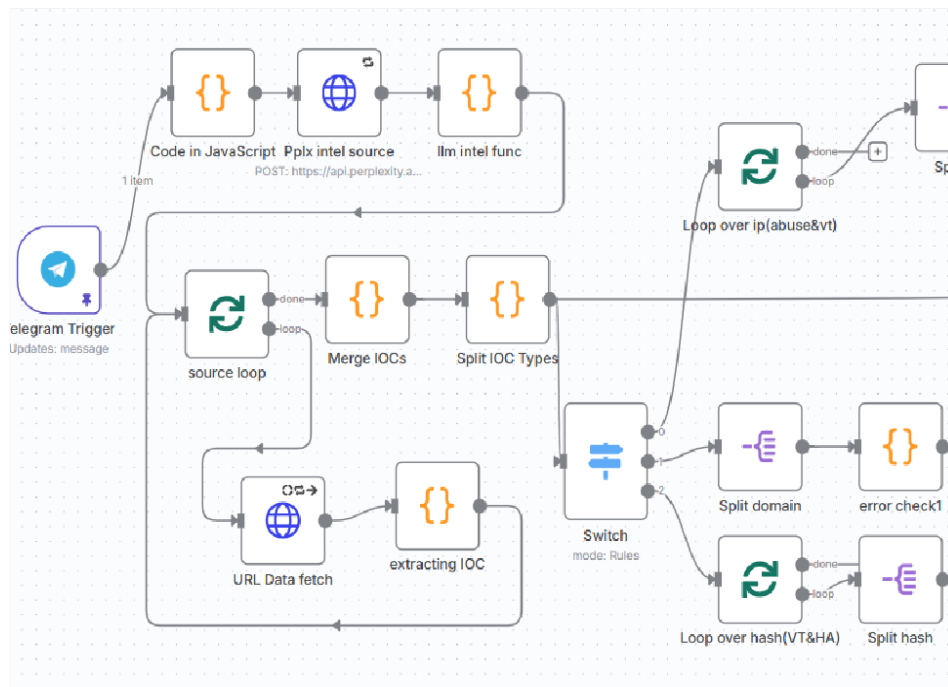


FIGURE 2. N8N Architecture

Throughout testing, several TI queries were run such as products “Fortinet 7.4”, “Microsoft Exchange 2024”, and “Apache Tomcat 9.0”. Each query returned a full report in about 30~40 seconds, these times were affected by network latency and response times from external APIs.

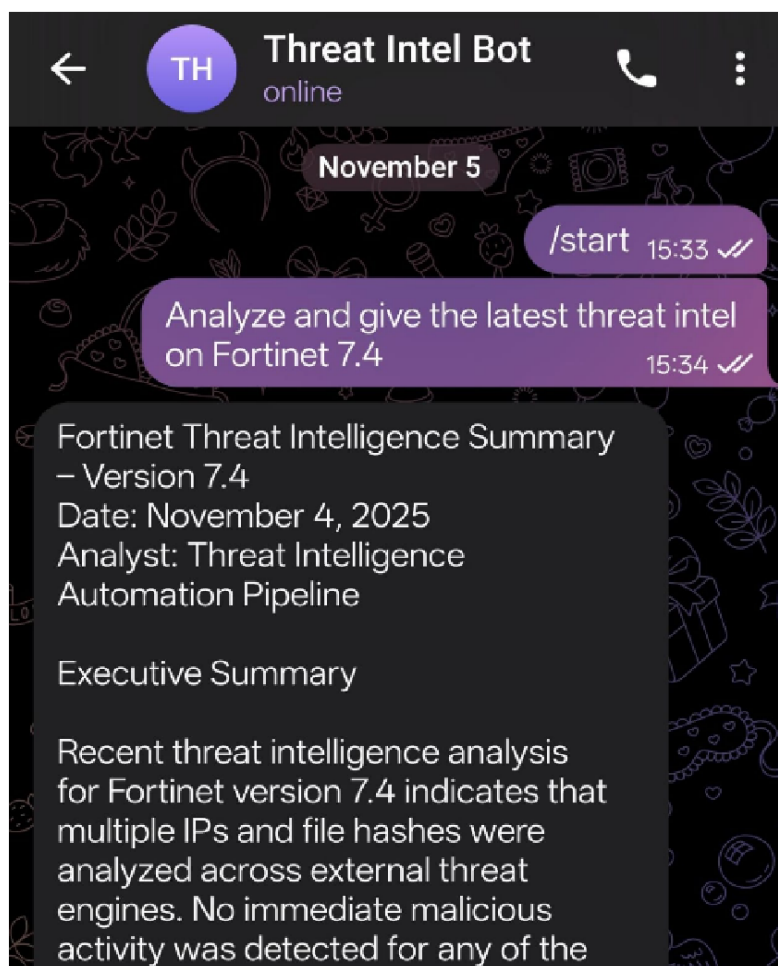


FIGURE 3. *Threat Intel Bot*

The Perplexity node always came back with recent citations from threats, so all the reports were backed by real and confirmable source URLs. The data enrichment from VirusTotal and AbuseIPDB added real-time context including malicious detection ratio, abuse confidence score, domain or ip reputation results.

Figure 3 depicts the last step of the workflow and is the result of the automation; this is the Telegram reply including the threat report in a structured way. The report consists of full sections including Executive Summary, Extracted IOCs, IOC Enrichment Results, Severity Classification, and Recommendations and have automated embedded citation links for traceability.

A sample system report for the query “Fortinet 7.4” is summarized below in Figure 4. The report has multiple sections which are auto generated using the Perplexity LLM and validated by enrichment data.

```

Threat Intelligence Summary
Product: Fortinet v7.4
Date: November 4, 2025

Executive Summary
Automated OSINT and enrichment analysis completed.
No immediate high-risk indicators detected.
Most IOCs belong to hosting or ISP infrastructure.

Extracted IOCs
CVEs: CVE-2025-34156, CVE-2025-34912
IPs: 43.228.217.173, 156.236.76.90
Domains: threatintelfeed.com
Hashes: 6e123e7f3202a8c1e9b1f94d8941580a25...

IOC Enrichment Summary

| Source         | Type | Verdict     |
|----------------|------|-------------|
| VirusTotal     | Hash | Clean       |
| AbuseIPDB      | IP   | Benign      |
| HybridAnalysis | Hash | No activity |

Observations
No confirmed malicious activity detected.
IPs used for scanning, benign behavior.

Recommendations
Monitor IPs and domains periodically.
Recheck IOCs weekly for status change.

Threat Level: LOW      Threat
Intel Sources Links
End of Automated Report.

```

FIGURE 4. *Threat Intel Report Format*

Figure 5 shows the final observations made from analysing the enriched IOCs as well as the recommended methods to handle it. At the end of the report all the threat intel sources used are given for further analysis and proof checking purposes.

The combination of every single node in n8n, from input processing to API enrichment and report generation was effective and stable on reiterated running. All of the intelligence cycle could be automated by avoiding the manual run of scripts with the modular node-based design, rendition-based configuration without the need for us to run the scripts manually.

Overall, the results show that the developed n8n-based pipeline integrates a large language model reasoning, multisource IOC enrichment, and structured reporting into a single automation workflow. It provides the analysts with full, referenced, formatted intelligence reports on demand through Telegram, minimizes manual efforts, and guarantees the latest product-specific threat intelligence visibility.

Older critical issues include ProxyLogon chain (CVE-2021-26855, -26858; SSRF/RCE, CVSS 9.8, widely exploited).⁵

Risk Assessment

- No active IOCs: Environment appears clean based on provided data.
- Patch Priority: Apply latest Cumulative Updates (e.g., CU15 with KB5071875/KB5066367) to mitigate 2025 CVEs. Unpatched systems vulnerable to spoofing/EoP leading to RCE.¹⁶⁷
-

RECOMMENDATIONS

: Verify patch level (e.g., Exchmem.dll version ≥15.2.1748.36), enable Extended Protection, monitor for MSRC indicators.¹⁴

THREAT LEVEL:  High

-  [Source 1](#)
-  [Source 2](#)
-  [Source 3](#)
-  [Source 4](#)
-  [Source 5](#)
-  [Source 6](#)
-  [Source 7](#)
-  [Source 8](#)

FIGURE 5. *Observations Delivered Via Telegram Report*

CONCLUSION

This study introduces a n8n-based fully automated threat intelligence and report generation pipeline. The framework incorporates Perplexity LLM for dynamic intelligence gathering based on OSINT and incorporates several enrichment sources such as VirusTotal, AbuseIPDB, and Hybrid Analysis to validate and analyze the trust level retrieved IOCs. The system automates all phases, from user query to report generation, in such a way that it can produce intelligence in real-time and for specific products using a very simple Telegram interface. The pipeline demonstrates a responsible use of large language models for contextual threat analysis that is factually grounded through multi-engine enrichment. The reports are structured and backed by citations, so they're immediately usable in operations, helping to increase both the pace and consistency of analyst workflows. This work provides a baseline for future research in AI orchestrated cyber threat intelligence automation and can be considered as a lightweight and scalable solution for modern security operations.

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